

Jason Zhang

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PROFESSIONAL SUMMARY

Data Scientist with expertise in financial modeling, risk stratification, and scalable ML systems. Proven track record at **Verizon** in driving multimillion-dollar margin improvements. Specialized in bridging the gap between deep technical architectures (LLMs, Time Series) and executive-level financial strategy.

TECHNICAL SKILLS

Languages/Libraries: Python, SQL, PyTorch, TensorFlow, Scikit-learn, Pandas, Flask, Java, TypeScript, R, Bash

Software/Tools: Git, Docker, Airflow, Spark, AWS, GCP, Mongo, Redis, Neo4j, ELK, Tableau, Jenkins

WORK EXPERIENCE

Data Scientist June 2025 - Present
Verizon Boston, MA

- Conducted breakeven analyses across **20+** risk segments, shaping promotion strategy and uncovering **multimillion-dollar** margin opportunities.
- Engineered **15+** predictive features from billing, tenure, and engagement data, improving classification confidence and risk stratification for collections payment modeling.
- Owned and automated reporting pipelines for **10+** consumer KPIs, reducing manual effort by **40%** and improving data accuracy for weekly executive reviews.
- Built and presented analytics narratives guiding nationwide promos, influencing leadership decisions and driving **multimillion-dollar** revenue protection and cost savings.

Data Scientist July 2024 - May 2025
U.S. DOT Volpe Center Cambridge, MA

- Engineered scalable ETL pipelines for safety incident data using Apache Airflow and AWS Glue, enabling real-time analysis of over **20,000** records annually, reducing data processing time by **60%**.
- Optimized a LightGBM model to predict offshore equipment failures and safety incidents, improving risk assessment accuracy by **40%**.
- Prototyped LLM RAG models using Hugging Face Transformers, increasing documentation retrieval efficiency by **30%** and saving an estimated **200** hours per year.

Software Engineer Co-op July 2023 - January 2024
Wolters Kluwer Waltham, MA

- Developed RESTful Spring Boot microservice for translation of medical content in UpToDate application using Google Cloud Translation, enabling processing of **100K+** medical documents daily.
- Remodeled user data schema, improving password encryption and enhancing protection for **2 million** users.

Data Science Co-op July 2022 - January 2023
MORSE Corp Cambridge, MA

- Developed a time series bootstrap model to estimate uncertainty in temporal data, improving predictive insights.
- Built a metadata-aware dataset split optimizer, reducing preprocessing time by **30%**.

EDUCATION

Northeastern University, Boston MA September 2024 - May 2026
Master of Science in Bioinformatics

Northeastern University, Boston MA September 2020 - May 2024
Bachelor of Science in Data Science and Mathematics **Honors:** Summa Cum Laude

APPENDIX

Data Scientist
U.S. DOT Volpe Center

July 2024 - May 2025
Cambridge, MA

- Engineered scalable ETL pipelines for safety incident data using **Apache Airflow** and **AWS Glue**, enabling real-time analysis of over **20,000 records** annually, reducing data processing time by **60%**.
- Optimized a **LightGBM** model to predict offshore equipment failures and safety incidents, improving risk assessment accuracy by **40%**.
- Prototyped **LLM RAG** models using **Hugging Face** Transformers, increasing documentation retrieval efficiency by **30%** and saving an estimated **200 hours** per year.
- Authored comprehensive memos on **NIST AI RMF**, driving **AI risk management adoption** on multiple teams.

Software Engineer Co-op
Wolters Kluwer

July 2023 - January 2024
Waltham, MA

- Developed RESTful Spring Boot microservice for translation of medical content in UpToDate application using **Google Cloud Translation**, enabling processing of **100K+ medical documents daily**.
- Re-implemented image retrieval backend using Elasticsearch, S3, and JAX-RS, reducing load time by **20%**.
- Remodeled user data schema, improving **password encryption** and enhancing protection for **2 million** users.

Data Engineering Researcher
Northeastern University Khoury College

December 2022 - May 2023
Boston, MA

- Re-architected database with Redis, accelerating speed of Evolutionary Computation framework by **20%** at scale.
- Leveraged Redis **distributed node clusters** and Redis Cloud for durability, achieving **99.99% uptime**.
- Improved codebase quality and maintainability by using **OOP best practices** and automated testing frameworks.

Data Science Co-op
MORSE Corp

July 2022 - January 2023
Cambridge, MA

- Developed a time series **bootstrap model** to estimate uncertainty in temporal data, improving predictive insights.
- Built a **metadata-aware** dataset split optimizer, reducing preprocessing time by **30%**.
- Streamlined model development by **40%** with custom analytics framework.
- Automated **ETL pipeline** optimizations using PySpark, reducing data lag by **15%**.
- Dockerized unit tests and automated documentation in a **CI/CD pipeline**, expediting release cycles by **10%**.

PROJECTS

AirflowIQ | *Python, Airflow, Spark, ARIMA, Docker*

August 2023 - Present

- A scalable, efficient data pipeline that delivers real time air quality insights around the world.

- Leverage big data tools for distributed data processing, resulting in faster analytics generation.
- Implement data quality monitoring using Airflow DAGs to ensure data integrity and provide timely alerts.
- Train ARIMA model to forecast ozone concentration and other relevant metrics for respiratory health.

Photo Quality Prediction | *Python, XGBoost, Tensorflow*

May 2023 - May 2023

- Feature engineered on complex multi-modal dataset to train XGBoost model predicting photograph downloads.
- Developed top 2 performing model in competition against 30 teams of Graduate Data Science students.
- Prototyped neural network architectures to compare against XGBoost model performance, balancing model complexity and explainability.

Twitter Dashboard | *Python, FastAPI, Typescript, React, MongoDB*

June 2022 - March 2023

- A dynamic Typescript React dashboard that generates analytics of users' Twitter data using sentiment analysis.
- Engineered ETL pipeline to store transformed Twitter API data in a custom MongoDB Schema, reducing data retrieval time by 50%.
- Performed testing and evaluation of pretrained sentiment analysis models to select the best candidate.

Instant Tutor Platform | *Javascript, Firebase, React*

March 2023 - July 2023

- An intuitive web application that connects students seeking academic assistance with rapid, high-quality tutoring services. Students submit problem images and receive video explanations from tutors with domain expertise.
- Supports secure user authentication and real-time chat.
- Designed Firestore database schema to handle complex user data.

Geeda (General EDA Framework Library) | *Python, Pandas, Pytest, Github Actions*

August 2022 - February 2023

- A general exploratory data analysis Python library to expedite the data science workflow. Increased personal EDA efficiency by 100%.
- Implement complete Github Actions CI/CD pipeline for rigorous testing and distribution.

Interests: Mandarin Language (Proficient), Gregorian Chant, Philosophy, Fitness, Skiing, Hiking, Sneaker Culture